

JESS Thermodynamic Database v8.9

Astatine

24-Jan-23

Reaction No. 71190							
$\text{At}_2 + 2\text{e}^{-1} = 2\text{At}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.3 (1SF)	5	CRV	6330

Reaction No. 74191							
$\text{H}+1_{\text{AtO}3-1} + 4\text{e}^{-1} + 4\text{H}+1 = \text{H}+1_{\text{AtO}-1} + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:1.4 (2SF)	3	CRV	22551

Reaction No. 74192							
$2\text{H}+1_{\text{AtO}-1} + 2\text{e}^{-1} + 2\text{H}+1 = \text{At}_2(\text{s}) + 2\text{H}_2\text{O}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.7 (1SF)	3	CRV	22551

Reaction No. 74193							
$\text{At}_2(\text{s}) + 2\text{e}^{-1} = 2\text{At}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.2 (1SF)	3	CRV	22551

Reaction No. 74194							
$\text{AtO}3-1 + 4\text{e}^{-1} + 2\text{H}_2\text{O} = \text{AtO}-1 + 4\text{OH}-1$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	E:0.5 (1SF)	3	CRV	22551

Reaction No. 79458							
$\text{At}(\text{s}) + \text{e}^{-1} = \text{At}^{-1}$							
No	t	I-Str	Medium	Value (Dev)	W	Tech.	Ref. #
1	25	0	Inf. Dilution	dG:-7.95 (3SF) kJ	5	CRV	7421

Data Assessment

Weight	Assessment
9	highest accuracy
8	multi-determined; excellent dependability
7	trustworthy; outstanding single determination
6	better than average reliability
5	average single determination
4	some doubt
3	poorly known
2	guess to achieve consistency (or just as bad)
1	no information but consistent
0	problematic/inconsistent

Techniques

Abbrev	Method
CRV	Critical review

References

Reference No.: 6330(95LiF)

CRC Handbook of Chemistry and Physics, 76th Edn., 1995 Lide DR; Frederikse HPR; A Ready-Reference Book of Chemical and Physical Data; CRC Press, Cleveland, Ohio, U.S.A.

Reference No.: 7421(97Mar)

Ion Properties, 1997 Marcus Y; Marcel Dekker Inc., New York, U.S.A.

Reference No.: 22551(06Inz)

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